Transport

25 questions: 8 multiple choice, 17 structured.

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1 · Part 4

Paper 1 & 2 Transport [1]

Amoeba is a single-celled organism that lives in pond water. It takes in oxygen and gets rid of carbon dioxide by diffusion across its cell membrane only.

Why can *Amoeba* rely on diffusion alone?

A It has a small surface area to volume ratio.

B It has a large surface area to volume ratio and short diffusion distances.

C Its cell membrane is fully permeable to all substances.

D It does not respire, so it needs very little oxygen.

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2 · Part 4

Paper 2 only Transport [1]

What is transpiration?

A the diffusion of carbon dioxide into a leaf through the stomata

B the release of oxygen from a leaf during photosynthesis

C the movement of water down the stem in the phloem

D the evaporation of water from the surface of a plant

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3 · Part 4

Paper 2 only Transport [1]

Which change in conditions would increase the rate of transpiration from a leafy shoot?

A an increase in wind speed

B an increase in humidity

C a decrease in temperature

D a decrease in light intensity

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4 · Part 4

Paper 2 only | Transport | [1]

Leila sets up a potometer to measure the rate of water uptake by a leafy shoot. She cuts the end of the stem while it is held under water, and she fits the stem into the potometer under water.

Why does she do this?

A to stop the leaves from losing water by transpiration

B to wash micro-organisms off the cut surface of the stem

C to stop air bubbles entering the stem and blocking water uptake

D to increase the rate of photosynthesis in the leaves

5 · Part 4

Paper 1 & 2 | Transport | [1]

Mature red blood cells have no nucleus.

How does this make red blood cells better suited to transporting oxygen?

A It leaves more space inside the cell for haemoglobin.

B It allows the cell to divide more rapidly to replace damaged cells.

C It allows the cell to carry out more aerobic respiration.

D It makes the cell membrane fully permeable to oxygen.

6 · Part 4

Paper 1 & 2 | Transport | [1]

Which blood vessel carries blood away from the right ventricle of the heart?

A aorta

B pulmonary vein

C pulmonary artery

D vena cava

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7 · Part 4

Paper 1 & 2 | Transport | [1]

Which feature of capillaries allows substances to be exchanged quickly between the blood and body cells?

A. They have a thick layer of muscle in their walls.

B. They have valves along their length.

C. They have a wide lumen.

D. They have walls that are one cell thick.

8 · Part 4

Paper 1 & 2 | Transport | [1]

Which blood vessel carries blood from the small intestine to the liver?

A. hepatic artery

B. hepatic portal vein

C. hepatic vein

D. vena cava

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9 · Part 4

Paper 1 & 2 Transport [6]

Scientists modelled organisms of different sizes as cubes. The table shows information about three cube-shaped models, P, Q and R.

surface area of a cube = $6 \times (\text{side length})^2$

volume of a cube = $(\text{side length})^3$

surface area to volume ratio = surface area ÷ volume

Model	Side length / mm	Surface area / mm^2	Volume / mm^3	Surface area to volume ratio
P	1	6	1	6.0
Q	2	24	8	3.0
R	5			

(a) Calculate the surface area to volume ratio of model R. Show your working. [3]

(b) Use the data in the table to explain why a large multicellular organism cannot rely on diffusion alone to supply oxygen to all of its cells. [3]

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10 · Part 4

Paper 1 & 2 Transport [6]

The table shows the time taken for oxygen to diffuse different distances through living tissue.

Distance / mm	Time taken / s
0.01	0.025
0.1	2.5
1	250
10	25 000

(a) Describe the relationship between the distance and the time taken for oxygen to diffuse. [2]

(b) Calculate the time taken, in hours, for oxygen to diffuse 10 mm through living tissue. (1 hour = 3600 s) [1]

(c) Some cells in a large animal are more than 10 mm from the body surface. A single-celled organism is about 0.1 mm across.

Explain why the large animal needs a transport system but the single-celled organism does not. Use the data in your answer. [3]

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11 · Part 4

Paper 1 & 2 | Transport | [6]

Sam stands a freshly cut celery stalk, with leaves at the top, in a beaker of water containing a red dye. The dye travels up the stalk in the xylem. After 3 hours, red lines first appear in the leaves at the top of the stalk, 24 cm above the cut end of the stalk.

(a) Describe the role of xylem in a plant. [3]

(b) Calculate the mean rate at which the dye moved up the stalk, in cm per hour.

rate = distance ÷ time [1]

(c) Phloem is the other transport tissue in the celery stalk. State two substances that are transported in the phloem. [2]

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12 · Part 4

Paper 1 & 2 Transport [6]

Rhys, a gardener, removes a complete ring of bark from the main stem of a young apple tree. This removes all of the phloem in a narrow band around the stem, but the xylem is not damaged.

Several weeks later, the gardener observes that:

- the stem just above the ring has swollen
- the leaves above the ring are green and have not wilted
- the roots have grown much more slowly than the roots of an undamaged tree.

(a) Explain why the leaves above the ring have not wilted. [2]

(b) Explain why the stem has swollen just above the ring. [2]

(c) Suggest why the roots have grown more slowly. [2]

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13 · Part 4

Paper 2 only

Transport

[7]

Root hair cells are found on the surface of young roots. Each root hair cell has a long, thin extension that grows out between soil particles. The vacuole of a root hair cell contains a more concentrated solution than the water in the soil.

(a) Explain how water moves from the soil into a root hair cell. [3]

(b) Explain how the shape of a root hair cell helps the plant to absorb water. [2]

(c) A field of crop plants is flooded with seawater. The water in the soil now contains a more concentrated solution than the vacuoles of the root hair cells.

Suggest why the crop plants wilt. [2]

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14 · Part 4

Paper 2 only

Transport

[7]

Four similar leafy shoots were each placed in a test tube of water. A layer of oil was poured onto the surface of the water in each test tube. Each test tube with its shoot was weighed, left in a different condition for 24 hours and then weighed again. All four shoots were kept in the same light intensity.

Shoot	Condition	Mass at start / g	Mass after 24 hours / g
W	still air, 20 °C	52.0	49.6
X	moving air from a fan, 20 °C	51.4	46.8
Y	still air, 30 °C	50.8	46.9
Z	still air, 20 °C, inside a closed, clear plastic box lined with wet paper towels	52.6	52.0

percentage loss in mass = (mass at start − mass after 24 hours) ÷ mass at start × 100

(a) Explain why a layer of oil was poured onto the surface of the water. [1]

(b) Calculate the percentage loss in mass for shoot X. Give your answer to 2 significant figures. [2]

(c) Explain why shoot Y lost more mass than shoot W. [2]

(d) Explain why shoot Z lost the least mass. [2]

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15 · Part 4

Paper 2 only

Transport

[9]

Kofi used a bubble potometer to investigate the effect of light intensity on the rate of water uptake by a leafy shoot. The potometer has a narrow glass tube with a millimetre scale. An air bubble in the tube moves along the scale as the shoot takes up water.

Kofi placed a lamp at different distances from the shoot. At each distance he measured how far the bubble moved in 5 minutes. He did three trials at each distance.

Distance of lamp from shoot / cm	Trial 1 / mm	Trial 2 / mm	Trial 3 / mm	Mean distance moved by bubble / mm
10	42	45	39	42
20	30	33	31	
40	18	16	20	18
80	9	11	10	10

(a) Calculate the mean distance moved by the bubble when the lamp was 20 cm from the shoot. [1]

(b) Calculate the rate of water uptake when the lamp was 10 cm from the shoot, in mm per minute.

rate = mean distance moved ÷ time [2]

(c) State two variables that Kofi should keep the same during this investigation. [2]

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(d) Explain why the rate of water uptake increased as the lamp was moved closer to the shoot. [3]

(e) The lamp also gives out heat. Suggest one change Kofi could make to his method so that only light intensity affects his results. [1]

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16 · Part 4

Paper 1 & 2 Transport [7]

In a hospital laboratory, a sample of human blood was mixed with a chemical that stops it clotting, and left to stand in a tube. After several hours the sample had separated. The top 55% of the sample was a pale yellow liquid. The cells had settled below it.

(a) State the name of the pale yellow liquid. [1]

(b) During exercise, muscles release heat energy. Explain how the pale yellow liquid helps to transfer this heat energy from the muscles to the skin. [2]

(c) The pale yellow liquid transports many substances around the body. Complete the table by stating the substance described in each row.

Description of substance carried	Substance
a waste product made in the liver and carried to the kidneys	
a waste gas carried from respiring cells to the lungs	
chemical messengers carried from glands to their target organs	
products of digestion carried from the small intestine to the liver	

[4]

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17 · Part 4

Paper 1 & 2 Transport [7]

At 4500 m above sea level, each breath of air contains less oxygen than a breath of the same volume at sea level. The table shows the mean number of red blood cells in the blood of two groups of adults.

Group	Mean number of red blood cells per mm^3 of blood
people living at sea level	5.0×10^6
people living at 4500 m above sea level	6.2×10^6

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(a) Calculate the percentage increase in the mean number of red blood cells for people living at 4500 m compared with people living at sea level.

percentage increase = (higher value − lower value) ÷ lower value × 100 [2]

(b) Suggest why it is an advantage for people living at 4500 m to have more red blood cells. [2]

(c) Red blood cells have a biconcave disc shape and contain haemoglobin.

Explain how each of these features makes red blood cells suited to transporting oxygen. [3]

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18 · Part 4

Paper 1 & 2 Transport [6]

Ana cut her hand on a rusty fence and bacteria entered the wound. Two types of white blood cell, phagocytes and lymphocytes, responded to the bacteria.

(a) Describe how phagocytes destroy the bacteria. [3]

(b) Lymphocytes released antibodies against the bacteria.

Explain why these antibodies would not protect Ana against a different type of bacterium. [2]

(c) A blood test two days later showed that Ana had more white blood cells per mm^3 of blood than normal. Suggest why. [1]

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19 · Part 4

Paper 2 only

Transport

[7]

On day 0, a person was given a vaccine against a disease. On day 60 the person was infected by the pathogen that causes the same disease. The concentration of antibodies against this pathogen in the person's blood was measured over 80 days.

Day	Antibody concentration / arbitrary units
0	0
5	2
10	10
15	18
20	12
30	4
60	2
63	40
66	150
70	200
80	160

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(a) Use the data to give two differences between the antibody response after the vaccination and the antibody response after the infection. [2]

(b) Calculate how many times greater the highest antibody concentration after the infection was than the highest antibody concentration after the vaccination.

number of times greater = highest concentration after infection ÷ highest concentration after vaccination [2]

(c) Explain why the antibody response after the infection was different from the response after the vaccination. [3]

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20 · Part 4

Paper 2 only

Transport

[6]

Tomas has a condition in which his blood contains far fewer platelets than normal.

(a) Explain why a cut on Tomas's skin bleeds for longer than a cut on the skin of a person with a normal number of platelets. [2]

(b) Suggest why Tomas is more likely than other people to get an infection in a cut. [2]

(c) Before treatment, Tomas had 20 000 platelets per mm^3 of blood. After a transfusion of platelets from a donor, he had 110 000 platelets per mm^3 of blood.

Calculate the percentage increase in Tomas's platelet count.

percentage increase = (value after − value before) ÷ value before × 100 [2]

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21 · Part 4

Paper 1 & 2 Transport [8]

The human heart has four chambers. Its walls are made of muscle. A wall of muscle called the septum separates the left side of the heart from the right side.

(a) Explain why the septum is important. [2]

(b) Explain why the wall of the left ventricle is thicker than the wall of the right ventricle. [2]

(c) Describe how blood is moved through the heart during one heartbeat, from when blood enters the atria until it leaves the ventricles. In your answer, include the role of the valves. [4]

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22 · Part 4

Paper 1 & 2 Transport [8]

Maria's heart rate was measured every 2 minutes. She rested, then ran on a treadmill for 6 minutes, then rested again.

Time / min	Activity	Heart rate / beats per minute
0	resting	68
2	running	110
4	running	142
6	running	150
8	resting	118
10	resting	86
12	resting	72

(a) Calculate the percentage increase in Maria's heart rate between 0 minutes and 6 minutes. Give your answer to 3 significant figures.

percentage increase = (higher value − lower value) ÷ lower value × 100 [2]

(b) Explain why Maria's heart rate increased while she was running. [4]

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(c) On a different day, just before the start of a race, Maria's heart rate was 90 beats per minute, although she was standing still. She felt nervous and excited.

Explain why her heart rate was higher than her normal resting heart rate. [2]

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23 · Part 4

Paper 1 & 2 Transport [8]

In a study, scientists followed a large number of men aged 50 to 59 years for ten years. The table shows the number of deaths from coronary heart disease (CHD) per year in four groups of these men.

Group	Deaths from CHD per 100 000 men per year
non-smokers with normal blood pressure	120
non-smokers with high blood pressure	260
smokers with normal blood pressure	300
smokers with high blood pressure	680

(a) Describe what the data show about the effects of smoking and high blood pressure on the risk of dying from CHD. [3]

(b) CHD develops when fatty material builds up in the walls of the coronary arteries.

Explain how this can cause a heart attack. [3]

(c) State two factors, other than smoking and high blood pressure, that increase the risk of developing CHD. [2]

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24 · Part 4

Paper 1 & 2 Transport [6]

Arteries and veins are two types of blood vessel.

(a) Arteries have thick walls containing muscle and elastic tissue.

Explain how this structure relates to the function of arteries. [3]

(b) Veins contain valves. Explain why veins need valves. [2]

(c) Veins have a wider lumen than arteries. Suggest one advantage of this to the body. [1]

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25 · Part 4

Paper 1 & 2 Transport [6]

Blood that has passed through the lungs travels to the kidneys, then returns to the heart.

(a) Name, in order, the blood vessels and the chambers of the heart that the blood passes through on its way from the lungs to the kidneys. [3]

(b) State the name of the blood vessel that carries blood away from the kidneys. [1]

(c) Explain why blood leaving the lungs contains more oxygen than blood arriving at the lungs. [2]

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